Bispecific Antibody (BsAb) Cytokine Release Syndrome (CRS), Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS)/Neurotoxicity and Supportive Care Guidance January 2026

Collaboration of VHA Oncology Field Advisory Board and the VHA National Formulary Committee

The following guidance is based on medical evidence, clinician input, and expert opinion. The content of the document is dynamic. It will be reviewed on a quarterly basis and revised as new information becomes available. The purpose of this document is to assist practitioners in clinical decision-making, to standardize and improve the quality of patient care, and to promote cost-effective drug prescribing. THE CLINICIAN SHOULD USE THIS GUIDANCE AND INTERPRET IT IN THE CLINICAL CONTEXT OF THE INDIVIDUAL PATIENT.

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Definitions

Bispecific Antibodies (BsAb): Contains 2 different target specific units in one antibody molecule. In oncology, these BsAbs generally link malignant cells to immune cells, blocking cell growth and activating the immune system (T cells).
Bispecific T-Cell Engager (BiTE): A BsAb that engages tumor-associated antigens on malignant cells and surface molecules on T-cells
Cytokines: Proteins that are produced and secreted by most cells that act as messengers facilitating multiple functions including the inflammatory response. Examples of cytokines includes interleukin, interferon, and tumor necrosis factor.
Cytokine Release Syndrome (CRS): An exaggerated systemic inflammatory response triggered by the effects of T-cell engaging therapies like BsAbs that cause release of inflammatory cytokines. CRS symptoms range from mild flu-like symptoms to severe multi-organ failure.
Infusion Related Reaction (IRR): A reaction related to drug hypersensitivity with symptoms experienced during the infusion or immediately following, typically within 6 hours of BiTE administration.
Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS): Neurologic toxicity caused by the inflammatory actions of cytokines released after BsAb therapy causing disruption of the blood-brain barrier and accumulation of inflammatory cytokines in the central nervous system. ICANS is a diagnosis of exclusion after other possibilities have been ruled out.

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BsAbs (BiTE) and Indications

Blinatumomab (BLINCYTO): Relapsed or Refractory CD19-positive B-cell precursor acute lymphoblastic leukemia (ALL)
Elranatamab (ELREXFIO): Relapsed or Refractory Multiple Myeloma (BCMA target) [REMS program]
Epcoritamab (EPKINLY): Relapsed or Refractory large B-cell lymphoma (CD20 target)
Glofitamab (COLUMVI): Relapsed or Refractory large B-cell lymphoma (CD20 target)
Mosunetuzumab (LONSUMIO): Relapsed or Refractory follicular lymphoma (CD20 target)
Talquetamab (TALVEY): Relapsed or Refractory multiple myeloma (GPRC5D target) [REMS program]
Tarlatamab (IMDELLTRA): Extensive-Stage Small Cell Lung Cancer after progression (DLL3 target)
Tebentafusp (KIMMTRAK): HLA-A*02:01 unresectable or metastatic uveal melanoma
Teclistamab (TECVAYLI): Relapsed or Refractory multiple myeloma (BCMA target) [REMS program]
Linvoseltamab (LYNOZYFIC): Relapsed or Refractory multiple myeloma (BCMA target) [REMS program]

Infusion-Related Reaction (IRR) Presentation, Diagnosis

An IRR is a reaction related to drug hypersensitivity with allergy-type symptoms experienced during or immediately following the BiTE infusion.

Symptoms associated with IRRs:
Anaphylaxis: sudden onset angioedema marked by difficulty breathing, stridor, wheeze, cyanosis or tachycardia, hypotension, dizziness, etc.
Systemic: chills, flushing, fatigue, hypotension, typically afebrile or low-grade fever
Cutaneous: urticaria and pruritis
Respiratory: cough and bronchospasm
Gastrointestinal: nausea and vomiting

- The earlier the reaction after drug exposure, the higher likelihood that it is an allergic or hypersensitivity reaction. A proposed timeframe in which to consider a reaction as infusion-related is one that occurs within 6 hours from the start of the BiTE infusion or within 2 hours of the end of the infusion (whichever is later). [11]
- Absence of a fever is typical of an IRR. [11]

Infusion-Related Reaction (IRR) Management

Anaphylaxis can be life-threatening and will require immediate action. Refer to local facility protocol for the management of anaphylaxis and other infusion-related reaction symptoms.

If no local protocol exists, consider the following management algorithm of IRRs after BiTE therapy.

	Grade 1	**Grade 2**	**Grade 3**	**Grade 4**
Symptoms & consequences	Moderate or symptoms not persistent. Low-moderate allergic symptoms: cough, chills, flushing, rash, pruritis, urticaria **Systemic intervention not indicated.**	Moderate and/or persistent symptoms, Use ABCDE* approach to assess severity of reaction. **Systemic intervention indicated**	Symptomatic bronchospasm; Allergy-related edema/ angioedema; hypotension, anaphylaxis symptoms **Urgent intervention indicated.**	Life-threatening consequences. **Urgent intervention indicated**
Timing of occurrence	Low to moderate reaction generally corresponds to reaction occurring after more than one hour of therapy		Severe reaction generally corresponds to reaction occurring rapidly after initiation of therapy (first minutes) and/or at beginning of second exposure to allergen	
Key points to management		Hold therapy until resolution of symptoms or return to Grade 1. Use H1/H2-receptor antagonists, corticosteroids; Bronchodilators and antileukotrienes for respiratory symptoms. Ensure sufficient IV fluids and oxygen support. When resuming therapy, consider restart at lower infusion rate and premedication. Closely monitor patients during resumption of therapy.		Stop BiTE infusion; if planning to resume therapy, consider risk vs. benefit, manage in coordination with ICU team and/or desensitization protocol
				Anaphylaxis: immediate epinephrine, fluid resuscitation, oxygen and ventilation support as needed in management with coordination of ICU team

*ABCDE Airway, Breathing, Circulation, Disability, Exposure

CRS Background

Basic CRS Pathophysiology

Cytokine Release Syndrome (CRS) is an exaggerated systemic inflammatory response due to the binding of BsAb to its antigen on the surface of target cells, causing activation of immune (e.g. T-cells) and non-immune cells that results in the immense release of inflammatory cytokines, fever and multiorgan dysfunction. Cytokines, such as interleukins, interferons, and tumor necrosis factor help to regulate immune responses, hematopoietic development, and response to infectious agents and inflammatory stimulation. Cytokines contribute to many of the clinical indicators of CRS. Symptoms may be progressive.

CRS Presentation, Diagnosis

I. **Clinical Presentation**

- **Time Course**- highest risk is **minutes to hours** after step-up or induction infusions, but it might occur once patient is discharged.

Bispecific	**Median Time to CRS onset after most recent dose**	**Median Duration of CRS**
Blinatumomab	2 days	5 days
Glofitamab	14 hours	2 days
Epcoritamab	24 hours	2 days
Tebentafusp	Day of infusion	2 days
Mosunetuzumab	4-46 hours	3 days
Teclistamab	2 days	2 days
Elranatamab	2 days	2 days
Talquetamab	27 hours	17 hours
Tarlatamab	13.5 hours	4 days
Linvoseltamab	11 hours	15 hours

- **Flu-like symptoms**: nausea, fatigue, headache, rash, diarrhea, arthralgia, myalgia
- **Fever** (≥38.0°C/>100.4°F) (not attributable to any other cause)
- **Hypotension**
- **Systemic inflammatory response syndrome** (circulatory collapse, vascular leakage, peripheral and/or pulmonary edema, renal failure, cardiac dysfunction, multiorgan failure)
- **Respiratory symptoms**: cough, tachypnea, hypoxia, ARDS
- **Rash and Urticaria** (allergic reaction)
- **Low-grade CRS is common and high-grade is rare**

II. **Differential Diagnoses**

- Infusion Related Reactions (IRRs) *refer to page 3*
- Tumor Lysis Syndrome (TLS): discriminate based on lab findings: hyperuricemia, hyperkalemia, hyperphosphatemia, hypocalcemia with TLS
- Infection, Sepsis: may coexist with CRS and simultaneous treatment for both may be considered
- Tumor progression

CRS Presentation, Diagnosis (continued)

III. **Evaluation and Diagnosis**-Unspecific symptoms and difficult to diagnose.

A. Evaluation

- Physical Examination- to include temperature, blood pressure, pulse oximetry or arterial blood gas (or mixed venous blood gas/O2 saturation), examination of skin, heart and lungs
- Labs- CBC with differential; Coagulation (PT/PTT, fibrinogen, fibrin D-dimer); Chemistry (serum electrolytes, kidney and liver function, uric acid, lactate, LDH); C-reactive protein and ferritin (inflammation); Microbiologic testing, especially in neutropenic patients (blood and urine cultures); cardiac markers as clinically indicated. Do not wait for lab results to initiate CRS management, if clinically indicated.
- Imaging: Chest X-ray, cross-sectional imaging (e.g., CT scan), or echocardiogram as clinically indicated.

B. **Lab findings**: cytopenias, elevated creatinine, elevated liver enzymes, dysregulated coagulation parameters, increased C-Reactive Protein

- CRS Management (see Management Section below) does not require lab testing and should not be delayed waiting for lab results

CRS Grading and Management

CRS Management by Grade-CRS is not well described with BsAb therapy; consensus guidelines from ASCTC for CRS with CAR-T therapy are generally utilized. In addition, there are consensus guidelines for CD3XCD20 BiTE therapy developed by a panel of experienced investigators and community practitioners.[12] Severity is determined by hypotension and hypoxia.[2]

- CRS with BsAbs is less severe than CRS with CAR T therapy.
- Early management of CRS may prevent escalation to a higher grade.
- For ALL Grades Hold BsAb therapy until CRS resolves then resume except for Grade 4
- If Outpatient, admit to inpatient if Grade >1 (Grade 1 might be managed as an outpatient). During initial step-up therapy, patient may need to be admitted to inpatient service depending on the BsAb.

CRS Grading and Management [1,2]

Grade	Management*
1 1. Temp >100.4F (not attributable to other causes) 2. Hypotension, Hypoxia: None	• Hold BsAb therapy until CRS resolves, then resume • Symptomatic/Supportive Care • Antipyretics-Acetaminophen 650mg orally Q6 hours prn • Consider infection work-up with blood and urine culture, chest radiography • IV hydration (for general support; if needed for hypotension treat as Grade 2) • If persistent or refractory fever to acetaminophen (<3 days), manage as Grade 2
2 1. Fever>/=100.4F 2. Hypotension responsive to fluid bolus. May use 1 low dose vasopressor 3. O2 needs <6L	• Hold BsAb therapy until CRS resolves, then resume • Continue Symptomatic/Supportive Care as for Grade 1 • Include IV fluid bolus and supplemental oxygen as needed; consider transfer to ICU if unresponsive to fluid bolus • If no improvement in hypotension after fluid bolus and/or hypoxia after supplemental oxygen, initiate dexamethasone 10mg IV every 8-12 hours. • If no improvement after 2-4 hours, provide tocilizumab 8mg/kg IV over 1 hour (not to exceed 800mg/dose); repeat every 8 hours until resolved to Grade 1. Limit to 3 doses per 24 hours with a maximum of 4 doses • If no improvement 24 hours after starting tocilizumab, manage as Grade 3
3 1. Fever >/=100.4 2. Hypotension requiring a vasopressor with or without vasopressin 3. O2 needs >6L (high-flow nasal canula, facemask, non-rebreather mask, or Venturi mask)	• Admit to ICU • Hold BsAb therapy until CRS resolves, then resume • Continue Supportive Care as in Grade 2 and add vasopressors if needed • If no improvement in hypotension after vasopressor and/or oxygen requirements, Initiate dexamethasone 10mg IV every 6 hours. • If no improvement after 2-4 hours, provide tocilizumab as per Grade 2 if maximum dose has not been reached and no improvement on high-dose steroids. If refractory, manage as Grade 4 • Continue dexamethasone for 3 days to resolution to ≤ grade 1, and rapidly taper when symptoms improve. • Echocardiogram of not already performed to assess cardiac function; begin hemodynamic monitoring
4 1. Fever>/=100.4 2. Hypotension requiring multiple vasopressors excluding vasopressin 3. Positive Pressure Ventilation (CPAP, BiPAP, Intubation, or mechanical intubation)	• Admit to ICU for hemodynamic monitoring, IV fluids, oxygen therapy, and vasopressors • Discontinue Bispecific Antibody • Continue Supportive Care as in Grade 3 plus mechanical ventilation as needed • Initiate dexamethasone 20mg IV every 6 hours until resolution to ≤ grade 1 followed by taper • Plus tocilizumab as per Grade 2 if maximum dose not reached • If no improvement on maximum doses of tocilizumab, consider other cytokine directed therapy (e.g. anakinra or siltuximab) for persistent grade 4 CRS despite maximal tocilizumab doses

* In patients receiving corticosteroids, consider antifungal prophylaxis

ICANS/Neurologic Toxicity Background

Pathophysiology

Similar to cytokine release syndrome (CRS), the pathophysiology of immune effector cell-associated neurotoxicity syndrome (ICANS) starts with the production of pro-inflammatory cytokines by chimeric antigen receptor (CAR) T cells and the activation of bystander immune cells such as macrophages in the tumor microenvironment.

Inflammatory cytokines and chemokines produced by CAR T cells and myeloid cells diffuse into the bloodstream and, eventually, result in disruption of the blood–brain barrier (BBB), with accumulation of cytokines and CAR T cells in the CNS together with activation of resident microglial cells.[8]

The neurologic toxicity (NT) of BsAb therapy has been described as ICANS, which was coined for the neurologic toxicity associated with CAR-T cell therapy. This description may not be accurate as the pathophysiology and clinical findings of BsAb-related vs. CAR-T cell therapy-related neurologic toxicity are not believed to be the same. In both settings, these toxicities are thought to be related to T-cell overactivation. Yet CAR-T cells are known to cross into the CNS, leading to increased protein and cytokine levels.[6] BsAb, especially those with the IgG backbone, are not expected to cross into the CNS. Therefore, neurologic adverse events within the BsAb clinical trial setting have been mild and rare events.

Although the neurotoxicity intensity and severity may vary among these immune therapies, there are similar management strategies. This document will focus on ICANS with noted differences that may exist for BsAb.

ICANS/Neurologic Toxicity Presentation, Diagnosis, Grading

I. **Clinical Presentation**

ICANS is typically associated with CRS, though few patients may develop ICANS without previous or concomitant CRS, ICANS may develop after CRS symptoms have resolved; rarely both complications can occur simultaneously. ICANS will typically manifest within 2-3 days of the onset of CRS. ICANS can present with varying degrees of neurologic changes. Initial presentation may be subtle with loss of attentiveness and language dysfunction. Other symptoms may include delirium, dysphasia, lethargy, difficulty concentrating, agitation, confusion, aphasia, depressed level of consciousness, encephalopathy, seizures, tremor, ataxia, loss of coordination or cerebral edema.

II. **Diagnosis**

- The diagnosis of ICANS should be considered while all other potential causes are being ruled out
- Suspect ICANS if new or worsening neurologic symptoms in the setting of recent immune effector cell (IEC), such as CAR-T cell therapy or BsAb therapy
- Initial presenting symptoms could be subtle as loss of attentiveness and/or speech dysfunction or tremors
- Additional investigation to r/o other potential causes should include review of concomitant medications or receipt of recent CNS-acting agents (i.e. opiates, benzodiazepines). Investigation may include head CT or brain MRI, lumbar puncture to r/o infectious causes

III. **Grading**

The ASTCT grading scale for CAR-T cell therapy-associated ICANS should also be used for BsAb therapy.

The process of grading ICANS involves evaluating the most severe symptom a patient has experienced, secondary to ICANS, among five domains: encephalopathy, level of consciousness, seizure, motor findings and elevated intracranial pressure/cerebral edema. Utilizing the ICE (**I**mmune effector **C**ell-associated **E**ncephalopathy) score, a 10-point encephalopathy assessment tool, patients are graded on the most severe symptom secondary to ICANS in five components: orientation, naming, following commands, writing and attention.

ICANS/Neurologic Toxicity Assessment, Management & Monitoring

I. Inpatient setting

A. Daily clinical assessment for ICANS/Neurologic Toxicity includes:

- Physical exam and review of vital signs
- Neurologic examination.
  - Initial signs may include subtle deficits in attention, changes in alertness and language
  - Neurologic assessment includes ICE score and neurotoxicity domain of ASTCT
  - Repeat at least twice daily and with any change in status
  - Consider Neurology consult
- Assess for signs of increased intracranial pressure. Consider Ophthalmology consult to perform fundoscopy
- Labs should include CBC, chem profile, PT, aPTT, fibrinogen, ferritin and CRP
- Monitor and correct severe hyponatremia
- Consider antiseizure prophylaxis, such as a non-sedating antiepileptic agent, for those at higher seizure risk, such as previous seizure history, CNS disease, EEG findings or brain lesions[2]
- Multidisciplinary discussion regarding clinical care setting (e.g. evaluate need for intensive clinical care)
  - Consider transfer to intensive care setting if progressive mental status changes, suspected cerebral edema, status epilepticus and/or patients with grade 3 or 4 neurotoxicity
  - Consider Neurology consult
  - Consider Ophthalmology consult to perform fundoscopy

B. Rule out other possible etiologies for neurologic changes. Additional evaluations may include:

- Concomitant medications
- Head CT or brain MRI for $\geq$ Grade 2 neurotoxicity
- Lumbar puncture to evaluate CSF to r/o suspicion of infection and $\geq$ Grade 3 neurotoxicity

ICANS/Neurologic Toxicity Assessment, Management & Monitoring (continued)

Management and Treatment

ASTCT ICANS Consensus Grading & Management by Grading for Adults

Neurotoxicity Grade	Grade 1	Grade 2	Grade 3	Grade 4
	ICE score* 7-9	**ICE score*** 3-6	**ICE score*** 0-2	**ICE score*** 0 (patient unarousable; unable to perform ICE)
	And/or depressed level of consciousness: Awakens spontaneously	**And/or depressed level of consciousness:** Awakens to voice	**And/or depressed level of consciousness:** Awakens only to tactile stimulus	**Depressed level of consciousness:** Unarousable or requires vigorous or repetitive tactile stimuli to arouse. Stupor or coma.
			Seizure: Any clinical seizure focal or generalized that resolves rapidly or nonconvulsive seizures on EEG that resolve with intervention	**Seizure:** Life-threatening prolonged seizure (> 5 min); or repetitive clinical or electrical seizures without return to baseline in between
			Elevated ICP/cerebral edema: Focal/local edema on neuroimaging	**Motor findings:** Deep focal motor weakness such as hemiparesis or paraparesis
				Diffuse cerebral edema on neuroimaging; decerebrate or decorticate posturing; or cranial nerve VI palsy; or papilledema; or Cushing's triad
Management of ICANS without CRS	Supportive care with IV hydration and aspiration precautions	Supportive care with IV hydration and aspiration precautions; Administer dexamethasone 10 mg IV x 2 (or equivalent) and reassess; Repeat every 6-12 hours if no improvement; Rapidly taper once symptoms improve to Gr 1	Transfer to ICU; Administer dexamethasone 10mg IV every 6-12 hours (or equivalent)	Transfer to ICU; Consider mechanical ventilation for airway protection; Administer high-dose methylprednisolone IV 1000 mg one to two times daily for 3 days; if not improvement, can increase to two to three times daily; Continue corticosteroids until Gr 1, then taper as appropriate
Management of ICANS with Concurrent CRS	Above care + Tocilizumab 8mg/kg IV over 1 hour. Repeat every 8 hours as needed. Limit to maximum of 3 doses/24-hr period	Above care + Transfer to ICU if also has Gr 2 CRS; Administer tocilizumab as Gr 1	Above care + Administer tocilizumab as Gr 1	Above care + Administer tocilizumab as Gr 1

*** ICE Scoring. Immune Effector Cell-Associated Encephalopathy (ICE) Assessment and Scoring**

Scoring Elements	Evaluation/Assessment	Score
Orientation	Patient is oriented to year, month, city, hospital	4 points
Naming	Ability to name/identify specific objects	3 points
Following Commands	Ability to follow simple commands	1 point
Writing	Ability to write a standard sentence	1 point
Attention	Ability to count backwards from 100 by 10	1 point

The sum of all elements totals the ICE score, a component of ASTCT ICANS Consensus Grading

Note that suggested management based upon recommendations for CAR-T cell therapy-associated ICANS

- Refer to prescribing information and clinical data for the BsAb product to ascertain potential risk for ICANS/neurotoxicity. Pay particular attention to:
  - Whether BsAb should be held, dose-reduced or discontinued
  - Supportive measures are recommended (i.e. IV hydration, corticosteroids)
  - Specialty consults suggested or transfer to intensive care unit

ICANS/Neurologic Toxicity Assessment, Management & Monitoring (continued)

II. Intensive Care setting

- Reserve intensive care setting for patients with worsening condition likely secondary to cerebral edema, status epilepticus and/or grade 3-4 ICANS/neurotoxicity. Patients will need close monitoring of neurologic, respiratory and cardiovascular functions.

III. Patient education on ICANS/Neurologic toxicity

- Discuss the possibility of ICANS/neurotoxicity associated with BsAb therapy
- Inform the patient/caregiver to be aware of signs and symptoms, which may include headache, confusion/ disorientation, attention loss/unable to focus, trouble speaking, sleepiness, problems with memory, dizziness, mental status changes.
- Advise patient and/or caregiver to immediately contact their provider should they experience any signs or symptoms that could indicate neurologic toxicity
- Instruct patient/caregiver to avoid driving or operating machinery if experiencing signs and symptoms related to neurotoxicity

Monitoring by Setting

Setting	Clinic/Outpatient	Inpatient and/or Intensive Care	Home, following discharge from either setting
Frequency	**Day of step-up doses & first full dose**	**Day of step-up doses, first full dose and subsequent days, as noted**	**Provide guidance to patient/caregiver prior to sending patient home**
IRR	Monitor for IRR symptoms likely to be seen w/in 6 hrs of start of dose or w/in 2 hours from end	Monitor for IRR symptoms likely to be seen w/in 6 hrs of infusion start or w/in 2 hours from end	Educate patients and caregivers to be cognizant of IRR symptoms on days an infusion was given
CRS	• VS • Weight • CBC with diff • CMP • Inflammatory markers: CRP and ferritin • Assess/grade CRS	• VS every 4 hours while awake (if stable) • Weight daily, I/O q 8 hrs • CBC with diff daily • CMP daily • Inflammatory markers: CRP and ferritin daily during step-up doses; continue daily until resolution • Coagulation parameters (PT, aPTT, fibrinogen) twice weekly • Eval/grade CRS q 12 hrs and whenever there is a change in clinical status; notify physician on call with any changes • Cardiac/hemodynamic monitoring by telemetry	• Provide medical team contact info and guidance on when to call • Instruct oral temperature to be taken each evening, unless clinical condition changes, then increase frequency to 3x/day • Encourage oral fluid intake (i.e. 1.5-2.0 L/day as tolerated)

ICANS		• Labs as noted above
		• Monitor & correct severe hyponatremia
		+ Neurologic assessment at least twice daily and with any change in status:
		• Rule out other possible etiologies for neurologic changes; additional evaluations may include:
		  ○ Concomitant meds
		  ○ Head CT or brain MRI for > Gr 2 neurotoxicity
		  ○ LP to r/o infection and > Gr 3 neurotoxicty
		• Initial signs of ICANS may include subtle deficit in attention, change in alertness and/or language
		• Grade with ICE score and neurotoxicity domain of ASTCT
		• Consider Neurology consult
		• Assess for signs of ICP
		• Consider Ophthalmology consult for fundoscopy
		• Consider antiseizure prophylaxis for those at higher risk
		• Multidisciplinary discussion regarding need for intensive care setting
			• Discuss the possibility of ICANS/neurotoxicity associated with BsAb therapy
			• Inform patient/caregiver to be aware of signs and symptoms, which may include headache, confusion/ disorientation, attention loss/unable to focus, trouble speaking, sleepiness, problems with memory, dizziness, mental status changes.
			• Advise patient and/or caregiver to immediately contact their provider should they experience any signs or symptoms that could indicate neurologic toxicity
			• Instruct patient/caregiver to avoid driving or operating machinery if experiencing signs and symptoms related to neurotoxicity

Communication Plan for Management of CRS and ICANS/Neurologic Toxicity

- Prior to patient admission, ensure communication with nursing, pharmacy, intensive care unit. Consider formation of a BiTE Therapy Team (physician, nurse, pharmacist) for improved care and decision-making.
- Consider notification to neurology service, as consult may be necessary.

I. Oncology Service

- ☐ Provider enters necessary forms (i.e. Prior Auth Drug Request and/or REMS forms)
- ☐ Orders entered for CRS and ICANS medications, including support medications need on PRN basis
- ☐ Notify inpatient medical team attending or service chief about plan to admit for CRS/ICANS management

II. Nursing

- ☐ Education for nurses (clinic and inpatient)
- ☐ Checklist for nursing (clinic and inpatient)
- ☐ REMS training and requirements met, if necessary (not every BsAb has a REMS)

III. Intensive care/step down units

- ☐ Notified of patient treatment on site (clinic or inpatient) with potential for CRS and/or ICANS
- ☐ Participate in multidisciplinary discussion for symptomatic patients
- ☐ Assist/manage transfer of care

IV. Pharmacy

- ☐ Education for pharmacists
- ☐ Checklist for pharmacy
- ☐ Support medication such as acetaminophen and corticosteroids should be available for immediate administration
- ☐ Necessary drugs in stock, including at least 2 doses of tocilizumab if available or can be obtained quickly
- ☐ Necessary forms completed (i.e. REMS)
- ☐ REMS training and requirements met, if necessary (not every BsAb has a REMS)

V. Inpatient Medicine Teams

- ☐ Consider education for medicine teams

VI. Patient /caregiver

- ☐ Educate on signs and symptoms of CRS
- ☐ Educate on signs and symptoms of ICANS/neurologic toxicity
- ☐ Provide Patient Wallet Card (pertaining to receipt of immune therapy) and/or Medication Guide from Prescribing Information

  Note: Patient Wallet Cards are not a part of FDA labeling for every BsAb

Additional Supportive Care

I. Cytopenias

All BiTEs are associated with varying degrees of hematologic toxicity, characterized by anemia, neutropenia and thrombocytopenia. The multiple myeloma BiTEs, particularly the BCMA-BiTEs, are associated with the highest rates of ≥ grade 3 cytopenias.

- Consensus guidelines recommend the use of colony-stimulating factors (CSFs) in patients with ≥ grade 3 neutropenia.[5]
- Caution that growth factors can induce or worsen CRS, therefore CSFs should not be given at the time of BiTE initiation; provide CSFs when the risk of CRS has subsided.
- Anemia should be managed per transfusion protocol.
- Thrombocytopenia should be managed per transfusion protocol. Retrospective and anecdotal evidence exists for the use of TPO-agonists in the CAR T-cell setting.[15, 16]

II. Infection management

Reported infection rates with BiTEs range from 15-45% depending on the agent. Risk factors leading to infection with BiTEs include: age, comorbidities, prior infections, CRS, lymphopenia, neutropenia, prolonged steroid use, hypogammaglobulinemia and refractory disease.

Hypogammaglobulinemia.

- Therapies that target B-cells and plasma cells will commonly cause hypogammaglobulinemia
- BCMA-targeted therapies are noted for significant B-cell depletion along with plasma cells
    - Teclistamab, for example, is responsible for IgG levels < 500 mg/dL in 75% of patients
- When to consider prophylactic administration of IVIG[17]
    - Patients with low levels of IgG (< 400mg/dL)
    - Patients with ≥ 2 severe recurrent infections by encapsulated bacteria, regardless of IgG level
    - Patients with life-threatening infection
    - Patients with documented bacterial infection with no or insufficient response to antibiotic therapy

Infection prevention and treatment

- Most common types of infection reported with BiTE therapy: respiratory/COVID, sepsis, CMV reactivation, PCP and fungal infections.
- Baseline screening should include: HIV, hepatitis B and C, CMV and EBV
- Vaccinations should be provided in advance of therapy, as antibody responses may be impaired from BiTEs
- Bacterial prophylaxis for those at high-risk of infection (i.e. recurrent bacterial infections, prolonged neutropenia, hypogammaglobulinemia)
- Antiviral prophylaxis against HSV and VZV in all patients
- Antifungal prophylaxis in patients with prolonged neutropenia or on corticosteroids for an extended duration
    - PCP prophylaxis should be considered, especially in those receiving BCMA-targeting BiTE therapies; the risk of grade ≥ 3 PCP infection is 3-4%
- Close monitoring for infections is essential to facilitate timely treatment.
- Management of suspected infections should follow established local protocols.

III. Keratotoxicity of GPRC5D-targeted therapies[5]

GPRC5D is not only expressed on myeloma cells, but also keratin-expressing cells. As such, patients may experience adverse effects affecting epidermal mouth and skin tissues, where keratinocytes are located.

Skin/dermatologic impact

- Erythema and rash occurred in 60% of talquetamab patients. Median time to onset was 25 days with a median time to improvement to ≤ grade 1 being 33 days.
- Supportive care includes the use of topical emollients.

Oral tissue impact

- Dysgeusia, dry mouth, dysphagia and stomatitis have occurred with talquetamab. 80% of patients experienced oral toxicities with a median time to onset of 15 days and median time to resolution of 43 days to baseline.

- Supportive care includes the use of saliva replacement therapy
    - VA National Formulary products include Artificial Saliva Oral Spray, Lozenge, Liquid and Gel.

Nursing Education / Checklist

Administration

Step-Up Dosing: Since CRS occurs most often in the first several days of BiTE therapy (but can occur at any time during therapy), the strategy of using stepwise increases during initial administration helps to decrease the risk of CRS. Package labels recommend step-up dosing in the inpatient setting due to long observation times for CRS or ICANS. CRS symptoms tend to subside with subsequent doses.

Pre-Medication & Supportive Meds: The use of pre-medications and supportive medications is drug-specific and is outlined in the package label. **Refer to Pages 13 and 18 for supportive medications, including premeds.** There may be differences in recommended pre-medications during initial step-up dosing, during subsequent cycles, and after the occurrence of CRS. In general, pre-medication falls into the following categories when recommended and should be given 60 minutes prior to the BsAb:

- Corticosteroid (usually prednisone or dexamethasone)
- Antihistamine (H1 antagonist) (usually diphenhydramine)
- Antipyretic (Acetaminophen)

Monitoring/Management of Toxicities

Outpatient/Clinic Setting

- ☐ Provide general supportive care and patient education
- ☐ Provide the patient with oncology contact information and guidance on when to call for symptoms
- ☐ Instruct the patient to measure oral temperature every evening. If the patient feels unwell or lethargic, the oral temperature should be measured three times a day. The patient should be instructed to seek medical attention for oral temperature ≥ 100.4°F and/or other symptoms concerning for CRS or ICANS.
- ☐ Encourage the patient to have adequate oral fluid intake
- ☐ Measure daily vital signs and weights
- ☐ Review the following labs during ramp up period:
    - Daily CBC with differential and complete metabolic profile
    - Coagulation parameters twice a week
    - C-Reactive Protein and ferritin daily during step-up and first dose, then as needed
- ☐ Assess CRS Grade and document in the medical record; reassess CRS grade if a change in status occurs (See Table in CRS Grading and Management Section)
- ☐ If within local facility nurse competency: Assess neurological status using ICE scale and level of consciousness and document in the medical record (See Table summarizing ICE scale in ICANS/Neurologic Toxicity Assessment, Management & Monitoring Section).
- ☐ Notify heme/onc provider immediately if patient experiences any neurologic or hemodynamic changes or changes in ICE score
    - RR <10 or >24 breaths per minute
    - Respiratory distress (cyanosis, wheezing)
    - SBP <90 mmHg
    - Temperature >38°C (100.4°F)
- ☐ Administer ordered corticosteroids or tocilizumab (Interleukin-6 inhibitor) therapy for CRS and/or ICANS.
- ☐ Facilitate transfer to inpatient or Intensive Care if needed for monitoring or progressing toxicity and provide sign-out to receiving inpatient or ICU team.

Nursing Education / Checklist (Continued)

Inpatient/Intensive Care Setting

- ☐ Provide general supportive care and patient education
- ☐ Perform vital signs every 4 hours
- ☐ Monitor and record IV and oral fluid intake and output
- ☐ Measure daily weights
- ☐ Review the following labs during ramp up period:
  - Daily CBC with differential and complete metabolic profile
  - Coagulation parameters twice a week
  - C-Reactive Protein and ferritin daily during step-up and first dose, then as needed
- ☐ Assess CRS Grade every 12 hours and document in the medical record; reassess CRS grade if a change in status occurs (See Table in CRS Grading and Management Section)
- ☐ If within local facility nurse competency: Assess neurological status using ICE scale and level of consciousness every 8-12 hours (or more frequently if there is a change) and document in the medical record (See Table summarizing ICE scale in ICANS/Neurologic Toxicity Assessment, Management & Monitoring Section).
- ☐ Notify <u>heme/onc and primary service teams</u> immediately if patient experiences any neurologic or hemodynamic changes or changes in ICE score
  - RR <10 or >24 breaths per minute
  - Respiratory distress (cyanosis, wheezing)
  - SBP <90 mmHg
  - Temperature >38ºC (100.4ºF)
  - Administer ordered corticosteroids or tocilizumab (Interleukin-6 inhibitor) therapy for CRS and/or ICANS.
- ☐ Facilitate transfer to Intensive Care if needed for intensive monitoring or progressing severe toxicity and provide sign-out to receiving ICU team.

Provider Education / Checklist

Monitoring/Management of Toxicities

Outpatient/Clinic Setting

- ☐ Provide general supportive care and patient education
- ☐ Provide the patient with oncology contact information and guidance on when to call for symptoms
- ☐ Instruct the patient to measure oral temperature every evening. If the patient feels unwell or lethargic, the oral temperature should be measured three times a day. The patient should be instructed to seek medical attention for oral temperature ≥ 100.4°F and/or other symptoms concerning for CRS or ICANS.
- ☐ Encourage the patient to have adequate oral fluid intake
- ☐ Review daily vital signs and weights
- ☐ Order and review the following labs during ramp up period:
    - Daily CBC with differential and complete metabolic profile
    - Coagulation parameters twice a week
    - C-Reactive Protein and ferritin daily during step-up and first dose, then as needed
- ☐ Review nurse documentation regarding CRS Grade. Assess CRS Grade and document in the medical record if not documented by nurse team; reassess CRS grade if a change in status occurs (See Table in CRS Grading and Management Section)
- ☐ Review nurse documentation regarding ICANS Grade if this is within local facility nurse competency. Assess neurological status using ICE scale and level of consciousness and document in the medical record if not documented by nurse team (See Table summarizing ICE scale in ICANS/Neurologic Toxicity Assessment, Management & Monitoring Section).
- ☐ Order corticosteroids or tocilizumab (Interleukin-6 inhibitor) therapy for CRS and/or ICANS based on above guidance document sections.
- ☐ Facilitate transfer to inpatient or Intensive Care if needed for monitoring or progressing toxicity, and provide sign-out to receiving inpatient or ICU team.

Inpatient/Intensive Care Setting

- ☐ Provide general supportive care and patient education
- ☐ Review every 4-hour vital signs
- ☐ Review IV and oral fluid intake and output
- ☐ Review daily weights
- ☐ Order and review the following labs during ramp up period:
    - Daily CBC with differential and complete metabolic profile
    - Coagulation parameters twice a week
    - C-Reactive Protein and ferritin daily during step-up and first dose, then as needed
- ☐ Review nurse documentation regarding CRS Grade. Assess CRS Grade every 12 hours and document in the medical record if not documented by nurse team; reassess CRS grade if a change in status occurs (See Table in CRS Grading and Management Section)
- ☐ Review nurse documentation regarding ICANS Grade if this is within local facility nurse competency. Assess neurological status using ICE scale and level of consciousness every 8-12 hours (or more frequently if there is a change) and document in the medical record if not documented by nurse team (See Table summarizing ICE scale in ICANS/Neurologic Toxicity Assessment, Management & Monitoring Section).
- ☐ Order corticosteroids or tocilizumab (Interleukin-6 inhibitor) therapy for CRS and/or ICANS.
- ☐ Facilitate transfer to Intensive Care if needed for intensive monitoring or progressing severe toxicity, and provide sign-out to receiving ICU team.

Pharmacy Checklist

I. Patient review

- ☐ Confirm name & identifier in CPRS
- ☐ Current age, height, weight, BSA
- ☐ Allergies, prior treatment-related reactions, comorbid conditions noted
- ☐ Diagnosis
- ☐ History of prior therapies, if appropriate

II. Treatment plan review

- ☐ Goal of therapy
- ☐ BsAb dose and regimen
- ☐ Duration of treatment/# cycles
- ☐ PADR adjudication
- ☐ Pertinent labs are current and within limits for treatment
- ☐ Anticipated start date of cycle #1, day #1
- ☐ Forecast date(s) of additional treatment days within a cycle and/or subsequent cycle

III. Drug order review

- ☐ Note facility area(s) where step-up and full BsAb dose(s) are planning to be administered (i.e. outpatient clinic vs. inpatient ward)
- ☐ Medication order for BsAb [will vary per BsAb; include dose, route, rate, etc.] step up and full doses
- ☐ Ensure appropriate supportive medications are ordered (see chart below)
- ☐ Antiseizure prophylaxis may be ordered based upon patient risk
- ☐ Note start date of cycle #1, day #1 and additional treatment days within cycle
- ☐ Note that not all BsAb start on cycle #1, day #1. For example, obinutuzumab is given cycle #1, day #1, followed by first dose of glofitamab on cycle #1, day #8 to deplete circulating and lymphoid tissue B cells
- ☐ If a dose is delayed, refer to prescribing information for recommendations to restart; review new orders to reflect these changes

Pharmacy Checklist (continued)

Supportive medications per FDA Package Labeling

Drug	***Pre-medications***	***Other meds***
Blinatumomab	Dexamethasone 16 mg IV	
Teclistamab	Dexamethasone 16mg PO/IV Diphenhydramine 50mg PO/IV Acetaminophen 650-1000mg PO	Antiviral prophylaxis
Glofitamab	Dexamethasone 20mg IV Diphenhydramine 50mg PO/IV Acetaminophen 500-1000 mg PO	Obinutuzumab day 1 Tumor lysis prophylaxis, if at risk Antiviral prophylaxis PJP prophylaxis
Epcoritamab	Dexamethasone 15mg PO/IV Diphenhydramine 50mg PO/IV Acetaminophen 650-1000mg PO	PJP prophylaxis Antiviral prophylaxis
Tebentafusp	None specified	
Mosunetuzumab	Dexamethasone 20mg IV Diphenhydramine 50-100 mg PO/IV Acetaminophen 500-1000 mg	
Elranatamab	Dexamethasone 20mg PO/IV Diphenhydramine 25mg PO Acetaminophen 650mg PO	PJP prophylaxis Antiviral prophylaxis
Talquetamab	Dexamethasone 16mg PO/IV Diphenhydramine 50mg PO Acetaminophen 650-1000mg PO	Antiviral prophylaxis
Tarlatamab	Dexamethasone 8mg IV or equivalent Cycle 1 1 liter Normal Saline over 4-5 hours Cycle 1	
Linvoseltamab	Dexamethasone 40mg IV Diphenhydramine 25mg PO/IV Acetaminophen 650-1000mg PO	

IV. Procurement process

- ☐ Review procedure to procure BsAb. Check PBM Formulary Management - Specialty Distribution Meds - All Documents (sharepoint.com)
- ☐ Ensure necessary forms are submitted, if appropriate (e.g. REMS)
- ☐ Confirm REMS Dispense Authorization (RDA) code, if necessary (e.g. teclistamab)
- ☐ Order BsAb in anticipation of identified start date
- ☐ Order ancillary medications needed to manage CRS
  - Tocilizumab 8 mg/kg IV over 1 hour (not to exceed 800 mg/dose). Repeat every 8 hours, prn. Limit to maximum of 3 doses/24-hour period and max of 4 doses total; ensure 24-hour supply on hand.
  - Dexamethasone 10 mg IV or equivalent
- ☐ All PADRs (Prior Auth Drug Requests) for BiTE therapies and their supportive meds (i.e. tocilizumab) should be submitted at one time to avoid any delays in obtaining supportive drugs.

V. BsAb preparation

- Review entire Preparation and Administration section of BsAb prescribing information
- Note dilution process may vary for each step-up dose, which may differ from the full dose
- Also note that within a BsAb product line preparation instructions may vary (e.g. epcoritamab 4 mg/0.8ml SDV must be diluted, but 48 mg/0.8ml SDV does not)
- Note storage and handling instructions of packaged product as well as diluted/reconstituted product

References

1. Lee DW, et al. ASCTC Consensus Grading for cytokine release syndrome neurologic toxicity associated with immune effector cells. Biol Blood Marrow Transplant 2019; 25: 625-638.
2. Santomasso BD, et al. Management of immune-related adverse events in patients treated with chimeric antigen receptor T-cell therapy: ASCO Guideline. J Clin Onc 2021; 39: 3978-3992.
3. IEC Therapy Toxicity Assessment and Management (also known as CARTOX) – adult. MD Anderson Cancer Center. V4 rev 9/2020
4. Common Terminology Criteria for Adverse Events (CTCAE), Version 5.0, November 2017, National Institutes of Health, National Cancer Institute. Available at: https://ctep.cancer.gov/protocoldevelopment/electronic_applications/docs/CTCAE_v5_Quick_Reference_8.5x11.pdf (Accessed March 27, 2018).
5. Ludwig H, Terpos E, van de Donk N, et al. Prevention and management of adverse events during treatment with bispecific antibodies and CAR-T cells in multiple myeloma: a consensus report of the European Myeloma Network. Lancet Oncol 2023; 24: e255-69.
6. Gust J, Finney OC, Li D, et al. Glial injury in neurotoxicity after pediatric CD19-directed chimeric antigen receptor T-cell therapy. Ann Neurol. 2019; 86: 42-54.
7. Morris EC, Neelapu SS, Giavridis T, Sadelain M. Cytokine release syndrome and associated neurotoxicity in cancer immunotherapy. Nat Rev Immunol 2022; 22: 85-96.
8. Anderson K, et al. Associated Toxicities-Assessment and Management Related to CAR T-cell therapy. Clin J Onc Nursing 2019; 23 (Supplement): 13-19.
9. Breslin S. Cytokine-Release Syndrome: Overview and Nursing Implications. Clin J Onc Nursing 2007; 11 (Supplement): 37-42.
10. Mucha SR, et al. Management and Prevention of Cellular-Therapy-Related Toxicity: Early and Late Complications. Curr Oncol 2023; 30: 5003-5023.
11. Geraud A, Hueso T, Laparra A, et al. Reactions and adverse Events Induced by T-cell Engagers as Anti-Cancer Immunotherapies, A Comprehensive Review. European Journal of Cancer 2024; doi.org/10.1016/k.ejca.2024.114075
12. Crombie JL, Graff T, Falchi L, et al. Consensus recommendations on the management of toxicity associated with CD3XCD20 bispecific antibody therapy. Blood 2024; 143: 1565-1575.
13. Browne EK, Daut E, Hente M, Turner K, Waters K, Duffy EA. Evidence-Based Recommendations for Nurse Monitoring and Management of Immunotherapy-Induced Cytokine Release Syndrome: A Systematic Review from the Children's Oncology Group. J Pediatric Oncology Nursing 2021; 38: 399-409.
14. Pagano L, Maschmeyer G, Lamoth F, et al. on behalf of ECIL. Primary antifungal prophylaxis in hematological malignancies. Updated clinical practice guidelines by the European Conference on Infections in Leukemia (ECIL). Leukemia 2025. 39: 1547-1557.
15. Mingot-Castellano ME, Reguera-Ortega JL, Zafra Torres D, et al. Use of eltrombopag to improve thrombocytopenia and transfusion requirement in anti-CD-19 CAR T-cell-treated patients. J Clin Med 2024; 13: 5117.
16. Beyar-Katz O, Perry C, et al. Thrombopoietin receptor agonist for treating bone marrow aplasia following anti-CD19 CAR T-cells, single-center experience. Ann Hematol 2022; 101: 1769-1776.
17. Raje N, Anderson K, Einsele H, et al. Monitoring, prophylaxis and treatment of infections in patients with MM receiving bispecific antibody therapy: consensus recommendations from an expert panel. Blood cancer Journal 2023; 13: 116.